

Course Code	Name of the Course				
216U03E613	Digital Image Processing				
Teaching Scheme	TH	P	TUT	Total	
	03	--	--	03	
Credits Assigned	03	--	--	03	
Evaluation Scheme	Marks				
	LAB/TUT CA	CA (TH)		ESE	Total
		IA	ISE		
	--	20	30	50	100

Course pre-requisites:

- Signals and Systems
- Digital Signal Processing
- Basics of Matrices

Course Objectives:

Digital Image Processing is one of the most widely investigated subjects under the domain of signal processing. The objective of the course is to introduce students the fundamentals of images and their processing using various techniques. The topics covered in this course find wide scope in applications such as photographic image processing, satellite imaging, bio-medical image processing, video codecs, etc. The course also aims to prepare students for further in-depth study of this field.

Course Outcomes (CO):

At the end of successful completion of the course the student will be able to

- CO1.** Understand theory and models in Image Processing
- CO2.** Interpret and analyze image enhancement algorithms in spatial domain
- CO3.** Interpret and analyze 2D signals in frequency domain through image transforms
- CO4.** Apply various methods for image segmentation and image description
- CO5.** Develop and design practical applications in various fields

Detailed Syllabus

Module No.	Unit No.	Details	Hrs	CO
1	Image Fundamentals		06	CO1
	1.1	Image acquisition, sampling and quantization, Spatial and Tonal resolution of images, types and formats of images, basic relationship between pixels, basic operations between images such as image addition, subtraction, blending and convolution. Color images: RGB and HSI models		
	1.2	# Self-Learning topics: Case study: Image fusion		
2	Image Enhancement in Spatial Domain		10	CO2
	2.1	Point Processing methods: Digital Negative, contrast stretching, thresholding, gray level slicing, bit plane slicing and log transform		
	2.2	Neighborhood Processing: Averaging filters, order statistics filters, high pass filters and high boost filters and their applications		
	2.3	Image Histogram stretching and equalization		
	2.4	Introduction to Image steganography, case studies on Processing DICOM files and Medical Imaging such as X ray, CT, MRI		
	2.5	# Self-Learning topics: Image zooming		
3	Frequency Domain Processing		10	CO3
	3.1	Two Dimensional Transforms: Concept of basis images, Discrete Fourier Transform (DFT), properties of DFT such as separability, frequency shift, average value and convolution, Discrete Cosine Transform, Hadamard Transform, Haar Transform		
	3.2	Frequency Domain filtering: DFT for filtering, Ideal, Gaussian, Butterworth filters for smoothening and sharpening, and Homomorphic filters.		
	3.3	Image Restoration: Degradation model, noise models, restoration using Wiener filters and Inverse filters		
		# Self learning topic: Walsh-Hadamard and KL Transform		
4	Image Segmentation, Description and Morphology		11	CO4
	4.1	Segmentation based on discontinuities: Point, line and edge detection.		
	4.2	Segmentation based on similarities and region based segmentation: Region growing, region splitting, region merging, split and merge		
	4.3	Boundary description: Chain codes, Shape number, Euler's number, Signatures, Fourier descriptors and statistical moments up to first order		

	4.4	Morphology: Dilation, erosion, opening, closing, hit or miss transform, thinning and thickening, and boundary extraction on binary images		
		# Self learning topic: Edge linking using Hough transforms.		
5	Image Compression		08	CO5
	5.1	Fundamentals of image compression and error criteria, definition of compression ratio, Mean Squared Error (MSE) and Peak Signal to Noise Ratio (PSNR)		
	5.2	Lossless Compression Techniques: Run length Encoding, Huffman codes, LZW technique		
	5.3	Lossy Compression Techniques: Improved Grey Scale Quantization, Transform coding		
		# Self learning topic: JPEG compression standard		
Total			45	

Reference Books*

Sr. No.	Name/s of Author/s	Title of Book	Publisher	Edition/Year
1.	Jayaraman, Esakkirajan and Veerakumar	<i>Digital Image Processing</i>	Tata McGraw Hill Education Pvt. Ltd., India	2 nd reprint, 2010
2.	Gonzalez and Woods	<i>Digital Image Processing</i>	Pearson Education, India	3 rd Edition, 2008
3.	Stan Birchfield	<i>Image Processing and Analysis</i>	Cenage Learning, USA	1 st Edition 2016
4.	Oge Marques	<i>Practical Image and Video Processing using MATLAB</i>	John Wiley & Sons Inc., US	1 st Edition, 2011

Learners should prepare all self-learning topics on their own. Self-learning topics will enable students to gain extended knowledge of the topic. Assessment of these topics may be included in IA.

* No questions expected on case study in ISE, IA and end semester theory examination.